

UNIT 1

Storyline

Human habitation and settlement has caused significant impacts on the natural environment. With increased population and continual development, ecosystems around the world have been changing at an alarming rate. Many of these impacts can be seen throughout Maryland and Montgomery County including our local watersheds. In this unit, students will study ecosystem biodiversity and the factors that shape populations, food webs, and energy flow. They will learn how organisms are classified as producers, consumers, or decomposers, how populations compete for limited resources, and how resource availability affects population growth within specific niches. Using models, students will explore the movement of matter and energy in ecosystems. The unit also connects learning to real life by examining the school community to evaluate human impacts on local ecosystems and aquatic habitats within Montgomery County and the Chesapeake Bay.

Anchoring Phenomenon

Many environmental changes can be observed in and around the Chesapeake Bay watershed. Students will explore how humans impact the water quality and environment around them and determine what actions they can take in order to lessen this impact.

Essential Questions:

- How does a system of living and non-living things operate to meet the needs of the organisms in an ecosystem?
- How do organisms interact with the living and nonliving environments to obtain matter and energy?
- What happens to ecosystems when the environment changes?
- How do matter and energy move through an ecosystem?

LS2.A: Interdependent Relationships in Ecosystems

- Organisms, and populations of organisms, are dependent on their environmental interactions both with other living things and with nonliving factors. (MS-LS2-1)
- In any ecosystem, organisms and populations with similar requirements for food, water, oxygen, or other resources may compete with each other for limited resources, access to which consequently constrains their growth and reproduction. (MS-LS2-1)
- Growth of organisms and population increases are limited by access to resources. (MS-LS2-1)
- Similarly, predatory interactions may reduce the number of organisms or eliminate whole populations of organisms. Mutually beneficial interactions, in contrast, may become so interdependent that each organism requires the other for survival. Although the species involved in these competitive, predatory, and mutually beneficial interactions vary across ecosystems, the patterns of interactions of organisms with their environments, both living and nonliving, are shared. (MS-LS2-2)

LS2.C: Ecosystem Dynamics, Functioning, and Resilience

- Ecosystems are dynamic in nature; their characteristics can vary over time. Disruptions to any physical or biological component of an ecosystem can lead to shifts in all its populations. (MS-LS2-4)
- Biodiversity describes the variety of species found in Earth's terrestrial and oceanic ecosystems. The completeness or integrity of an ecosystem's biodiversity is often used as a measure of its health. (MS-LS2-5)

ESS3.C: Human Impacts on Earth Systems

- Human activities have significantly altered the biosphere, sometimes damaging or destroying natural habitats and causing the extinction of other species. But changes to Earth's environments can have different impacts (negative and positive) for different living things. (MS-ESS3-3)

Outdoor ed

Outdoor Environmental Education: Grade 6 Residential Program

	Sixth graders at each middle school in MCPS participate in a three-day, two-night residential outdoor environmental education program. While in residence, students study various aspects of the local environment as they participate in outdoor investigations that are directly connected to the grade six curriculum and the MSDE environmental education standards. Students are accompanied by their teachers, who, in collaboration with an OEEP staff member, provide instruction and supervision during their stay.
Curriculum	During their stay, students participate in a variety of lessons and activities that support the Montgomery County Public Schools curriculum. Instruction focuses on environmental studies and science, while integrating math, reading, world studies, and physical education. Each school determines its own instructional schedule and classes. Generally, students will participate in four to five of the lessons listed below as well as evening activities. Classes are led by staff from the school and the outdoor environmental educator.
Teacher Resources:	6th Grade Residential Program Website Links to an external site.
Student Service Learning	As part of the Investigations in Earth Science and Outdoor Environmental Education curriculum, students earn 10 hours for the successful completion of an environmental Student Service Learning project when they pass grade 6 science. This project is facilitated by the school and may include a home or school study of current resource usage followed by a 30-Day Intent to Act Pledge where students

	develop a plan that reduces their environmental impact to become more climate neutral.
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***Please note: Attending Outdoor Education does not fulfill the requirement for earning the 10 SSL hours. In order to earn these hours, students must complete the companion project or a related project outside of the Outdoor Education Experience.

Investigations in Earth Science Vocabulary List

A

Abiotic Factor: A nonliving physical or chemical part of an ecosystem. (p. 474) active transport The process of using energy to move materials through a membrane.

Abrasion: The process of wearing something down by friction.

Absolute age: The actual age in years of an event or object.

Acid Rain: Rain that has become more acidic than normal due to pollution.

Aftershock: A smaller earthquake that follows a more powerful earthquake in the same area.

Air Mass: A large volume of air that has nearly the same temperature and humidity at different locations at the same altitude.

Air Pollution: Harmful materials added to the air that can cause damage to living things and the environment.

Air Pressure: The force of air molecules pushing on an area.

Altitude: The distance above sea level.

Asthenosphere: The layer in Earth’s upper mantle and directly under the lithosphere in which rock is soft and weak because it is close to melting.

Atmosphere: The outer layer of gasses of a large body in space, such as a planet or star; the mixture of gasses that surrounds the solid Earth; one of the four parts of the Earth system.

Axis of Rotation: An imaginary line about which a turning body, such as Earth rotates.

B

Barometer: An instrument that measures air pressure in the atmosphere.

Barrier island: A long, narrow island that develops parallel to a coast as a sandbar builds up above the water’s surface.

Biodiversity: The number and variety of living things found on Earth or within an ecosystem.

Biomass: Organic matter that contains stored energy from sunlight and that can be burned as fuel.

Biome: A region of Earth that has a particular climate and certain types of plants. Examples are tundra, taiga, desert, grassland, temperate and tropical forests. (p. 494) **biotic factor** (by-AHT-i h k) A living thing in an ecosystem. (p. 494) **blood** A fluid in the body that delivers oxygen and other materials to cells and removes carbon dioxide and other wastes. (p. 641) **blubber** A layer of fat in some sea mammals that lies beneath the skin. It insulates the animal from cold and stores reserve energy. (p. 456) **budding** A process of asexual reproduction in which an organism develops as an outgrowth of the parent. Each bud can grow into a new organism, breaking free and becoming separate and independent. (p. 89) **c** **capillary** A narrow blood vessel that connects arteries with veins. (p. 645) **carbohydrate** (KAHR- boh-HY-drayt) A type of molecule made up of subunits of sugars and used for energy and structure. (p. 42)

Biosphere: All living organisms on Earth in the air, on the land, and in the waters; one of the four parts of the Earth system.

Biotic Factor: A living thing in an ecosystem.

Blizzard: A blinding snowstorm with winds of at least 56 kilometers per hour (35 mi/h), usually with temperatures below -7°C (20°F).

C

Carrying Capacity: The maximum size that a population can reach in an ecosystem. (pp. 242, 529) **cell** The smallest unit that is able to perform the basic functions of life.

Chemical Weathering: The breakdown or decomposition of rock that takes place when minerals change through chemical processes.

Cleavage: The property of a mineral that describes its tendency to break along flat surfaces. .

Community: All the populations that live and interact with each other in a particular place. The community can live in a place as small as a pond or a park, or it can live in a place as large as a rain forest or the ocean.

Competition: The struggle between two or more living things that depend on the same limited resource.

Condensation: The process by which water vapor (water in its gas form) turns into liquid.

Conduction: The transfer of heat/energy through direct contact

Conservation: The process of saving or protecting a natural resource.

Convection: The transfer of heat/energy through the movement of a fluid like air, molten rock, or water.

- In Earth's mantle, convection is thought to transfer energy by the motion of solid rock, which when under great heat and pressure can move like a liquid.
- In the Atmosphere, convection circulates air throughout the atmosphere as warm air rises from the surface and cool air moves in to take its place causing wind.

Convection Current: A circulation pattern in which material is heated and rises in one area, then cools and sinks in another area, flowing in a continuous loop.

Convergent Boundary: A boundary along which two tectonic plates push together, characterized either by subduction or a continental collision.

Coriolis Effect: The influence of Earth's rotation on objects that move over Earth. (p. 545) **corona** The outer layer of the Sun's atmosphere.

Crust: A thin outer layer of rock above a planet's mantle, including all dry land and ocean basins. Earth's continental crust is 40 kilometers thick on average and oceanic crust is 7 kilometers thick on average.

Crystal: A solid substance in which the atoms are arranged in an orderly, repeating, three-dimensional pattern.

D

Dam: A structure that holds back and controls the flow of water in a river or other body of water.

Data: Information gathered by observation or experimentation that can be used in calculating or reasoning. Data is a plural word; the singular is datum.

Density: A property of matter representing the mass per unit volume.

Deposition: The process in which transported sediment is laid down. (p. 145)

Desalination: The process of removing salt from ocean water. Desalination is used to obtain fresh water.

Desertification: The expansion of desert conditions in areas where the natural plant cover has been destroyed.

Dew point: The temperature at which air with a given amount of water vapor will reach saturation.

Divergent Boundary: A boundary along which two tectonic plates move apart, characterized by either a mid-ocean ridge or a continental rift valley.

Drought: A long period of abnormally low amounts of rainfall.

Dune: A mound of sand built up by wind. (p. 161)

E

Earthquake: A sudden violent shaking or movement of the land due to a slip along a fault or plate boundary.

Eclipse: An event during which one object in space casts a shadow onto another. On Earth, a lunar eclipse occurs when the Moon moves through Earth's shadow, and a solar eclipse occurs when the Moon's shadow crosses Earth

Ecology: The scientific study of how living things interact with each other and their environment.

Ecosystem: All the living and nonliving things that interact in a particular environment. An ecosystem can be as small as a meadow or a swamp or as large as a forest or a desert.

Electromagnetic Radiation: Energy that travels across distances as certain types of waves. Types of electromagnetic radiation are radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, x-rays, and gamma rays.

Elevation: A measure of how high something is above a reference point, such as sea level.

El Nino: A disturbance of wind patterns and ocean currents in the Pacific Ocean that causes temporary climate changes in many parts of the world.

Energy: The ability to do work or to cause a change.

Environment: Everything that surrounds a living thing. An environment is made up of both living and nonliving factors.

Epicenter: The point on Earth's surface directly above the focus of an earthquake.

Equator: An imaginary east-west line around the center of Earth that divides the planet into the Northern Hemisphere and the Southern Hemisphere; a line set at 0° latitude. (p. 18)

Erosion: The process in which sediment is picked up and moved from one place to another.

Estuary: A shoreline area where fresh water from a river mixes with salt water from the ocean.

Evaporation: A process by which a substance changes from its liquid state to its gas state by random particle movement. Evaporation usually occurs at the surface of a liquid over a wide range of temperatures.

Extinction: The permanent disappearance of a species.

Extrusive, Igneous Rock: Igneous rock that forms as lava cools on the Earth's surface.

F

Fault: A fracture in Earth's lithosphere along which blocks of rock move past each other.

Fault-block mountain: A mountain that forms as blocks of rock move up or down along normal faults in areas where the lithosphere is being pulled apart.

Felsic: Igneous rocks made up of high silica, light colored minerals.

Focus: In an earthquake, the point underground where the rocks first begin to move.

Folded Mountain: A mountain that forms as continental crust is compressed and rocks bend into large folds.

Foliation: The arrangement of minerals within rocks into flat or wavy parallel bands; a characteristic of most metamorphic rocks.

Fossil: A trace or the remains of a once-living thing from long ago.

Fossilization: The processes that lead to the preservation of plant and animal remains over time

Fossil Fuels: Fuels formed from the remains of prehistoric organisms that are burned for energy.

Fracture: The tendency of a mineral to break into irregular pieces.

Freezing: The process by which a substance changes from its liquid state into its solid state.

Freezing Point: The temperature at which a substance changes from its liquid state to its solid state through freezing.

Freezing Rain: Rain that freezes when it hits the ground or another surface and coats the surface with ice.

Fresh Water: Water that is not salty and has little or no taste, color, or smell. Most lakes and rivers are made up of fresh water.

Front or Frontal Boundary: The boundary between air masses.

G

Gas: A state of matter different from liquid and solid, with no definite volume and no definite shape.

Generator: A device that converts kinetic energy, or the energy of motion, into electrical energy. Generators produce electric current by rotating a magnet within a coil of wire or rotating a coil of wire within a magnetic field.

Geologic Time Scale: The summary of Earth's history, divided into intervals of time defined by major events or changes on Earth.

Geosphere: All the features on Earth's surface-continents, islands, and seafloor-and everything below the surface-the inner and outer core and the mantle; one of the four parts of the Earth system.

Geothermal Energy: Heat energy that originates from within Earth and drives the movement of Earth's tectonic plates. Geothermal energy can be used to generate electricity.

Glacier: A large mass of ice that exists year-round and moves over land.

Global Winds: Winds that travel long distances in steady patterns over several weeks. (p. 544) gravity The force that objects exert on each other because of their mass.

Gravity: The force that objects exert on each other because of their mass.

Greenhouse Effect: The process by which certain gasses in a planet's atmosphere absorb and emit infrared radiation, resulting in an increase in surface temperature.

Greenhouse Gasses: Gasses, such as carbon dioxide and methane, that absorb and give off infrared radiation as part of the greenhouse effect.

Groundwater: Water that collects and is stored underground.

H

Habitat: The natural environment in which a living thing gets all that it needs to live; examples include a desert, a coral reef, and a freshwater lake.

Hail: Layered lumps or balls of ice that fall from cumulonimbus clouds due circulation within the cloud system.

Half-life: The length of time it takes for half of the atoms in a sample of a radioactive element to change from an unstable form into another form.

Hardness: The resistance of a mineral or other material to being scratched.

High-pressure System: A generally calm and clear weather system that occurs when air sinks down in a high-pressure center and spreads out toward areas of lower pressure as it nears the ground.

Hot Spot: An area where a column of hot material rises from deep within a planet's mantle and heats the lithosphere above it, often causing volcanic activity at the surface.

Humidity: The amount of water vapor in air.

Humus: The decayed organic matter in soil.

Hurricane: A tropical low-pressure system with sustained winds of 120 kilometers per hour (74 mi/h) or more.

Hydroelectric Energy: Electricity that is generated by the conversion of the energy of moving water.

Hydrogen Fuel Cell: A device that uses hydrogen and oxygen to produce electricity. The byproducts are heat and water.

Hydrosphere: All water on Earth-in the atmosphere and in the oceans, lakes, glaciers, rivers, streams, and underground reservoirs; one of the four parts of the Earth system.

Hydrothermal Vent: An opening in the sea floor from which heated water rises and mixes with the ocean water above.

Hypothesis: A tentative explanation for an observation or phenomenon. A hypothesis is used to make testable predictions.

I

Ice Age: A period of time during which surface temperatures drop significantly and huge ice sheets spread out beyond the polar regions.

Iceberg: A mass of floating ice that broke away from a glacier.

Igneous Rock: Rock that forms as molten rock cools and becomes solid.

Impermeable: Resistant to the passage of water.

Index Fossil: A fossil of an organism that was common, lived in many areas, and existed only during a certain span of time. Index fossils are used to help determine the age of rock layers.

Infrared Radiation: Radiation of lower frequencies than visible light.

Inner Core: A solid sphere of metal, mainly nickel and iron, at Earth's center

Intertidal Zone: The narrow ocean margin between the high-tide mark and the low-tide mark.

Intrusive Igneous Rock: Igneous rock that forms as magma cools below Earth's surface.

Invertebrate: An animal that has no backbone.

Island Arc: A chain of islands formed from volcanic activity due to subduction along an ocean-ocean convergent boundary.

Isobar: A line on a weather map connecting places that have the same air pressure.

Isotherm: A line on a weather map connecting places that have the same temperature.

J

Jet Stream: A wind that flows in the upper troposphere from west to east over vast distances at great speeds.

L

Lahar: A fast moving mudflow or debris flow composed of a mixture of pyroclastic material, rocky debris and water.

Latitude: The distance in degrees north or south from the equator.

Lava: Molten rock that reaches a planet's surface through a volcano.

Lava Flow: Streams of molten rock that pour or ooze from an erupting vent.

Limiting Factor: A factor or condition that prevents the continuing growth of a population in an ecosystem.

Liquefaction: A process in which the shaking of ground causes loose, wet soil to act like a liquid.

Lithosphere: The layer of Earth made up of the crust and the rigid rock of the upper mantle, averaging about 40 kilometers thick and broken into tectonic plates.

Loess: Deposits of fine-grained, wind-blown sediment.

Longitude: The distance in degrees east or west of the prime meridian. Longitude lines are numbered from 0° to 180°.

Low-pressure System: A large and often stormy weather system that occurs when air moves around and into a low-pressure center, then moves up to higher altitudes.

Luster: The property of a mineral that describes the way in which light reflects from its surface. Major types of luster are metallic and nonmetallic.

M

Macroinvertebrate: Insects in their nymph and larval stages, snails, worms, crayfish, and clams that spend at least part of their lives in water and are important in determining the health of a local watershed.

Mafic: Igneous rocks made primarily of iron rich, dark colored minerals.

Magma: Molten rock beneath Earth's surface.

Magnetic Reversal: A switch in the direction of Earth's magnetic field so that the magnetic north pole becomes the magnetic south pole and the magnetic south pole becomes the magnetic north pole.

Magnetic Stripe: Patterns of magnetism created by ocean floor rocks recording changes in Earth's magnetic field over time.

Mantle: The layer of rock between Earth's outer core and crust, in which most rock is hot enough to flow in convection currents; Earth's thickest layer.

Marine Climate: A climate influenced by a nearby ocean, with generally mild temperatures and steady precipitation.

Mass: A measure of how much matter an object is made of.

Mass Extinction: One of several periods in Earth's history when large numbers of species became extinct at nearly the same time.

Mass Wasting: The downhill movement of loose rock or soil.

Matter: Anything that has mass and volume. Matter exists ordinarily as a solid, a liquid, or a gas.

Mechanical Weathering: The breakdown of rock into smaller pieces of the same material without any change in its composition.

Mercalli Scale: The measurement scale measures the intensity of an earthquake.

Metamorphic Rock: Rock formed as heat or pressure causes existing rock to change in structure, texture, or mineral composition.

Metamorphism: The process by which a rock's structure or mineral composition is changed by pressure or heat.

Meteorologist: A scientist who studies weather.

Mid-ocean Ridge: A long line of sea-floor mountains where new ocean crust is formed by volcanic activity along a divergent boundary.

Mineral: A substance that forms in nature, is a solid, has a definite chemical makeup, and has a crystal structure.

Monsoon: A wind that changes direction with the seasons.

N

Natural Resource: Any natural material or energy from Earth's environment that humans use to meet their needs.

Neap Tide: A tide of small range occurring during the first- and third-quarter phases of the Moon.

Nebula: A cloud of gas and dust in space. Stars form in nebulae.

Neutron Star: A dense core that may be left behind after a higher-mass star explodes in a supernova.

Nonpoint-source Pollution: Pollution with a source that is hard to find or scattered.

Nonrenewable Resource: A resource that exists in a fixed amount or is used up more quickly than it can be replaced in nature.

Nuclear Fission: The process of splitting the nuclei of radioactive atoms, which releases huge amounts of energy mainly in the form of radiation and heat energy.

O

Ocean Current: A mass of moving ocean water.

Oceanic-Continental Subduction: A boundary along which a plate carrying oceanic crust sinks beneath a plate with continental crust. This type of subduction typically forms volcanoes on the continents above the zone.

Oceanic-Oceanic Subduction: A boundary along which a plate carrying oceanic crust sinks beneath another plate with oceanic crust. This type of subduction typically forms island arcs.

Organism: An individual living thing, made up of one or many cells, that is capable of growing and reproducing.

Outer Core: A layer of molten metal, mainly nickel and iron, that surrounds Earth's inner core. (p. 186)

overfishing The catching of fish at a faster rate than they can reproduce.

Ozone: A gas molecule that consists of three oxygen atoms.

P

Paleontology: the branch of science concerned with the fossil remains of plants and animals.

Pangaea: A hypothetical supercontinent that included all of the landmasses on Earth. It began breaking apart about 200 million years ago.

Particulates: Tiny particles or droplets, such as dust, dirt, and pollen, that are mixed in with air.

Permeable: Allowing the passage of water.

Pillow Basalt: A volcanic igneous rock that forms when lava of basaltic composition is erupted underwater.

Point-source Pollution: Pollution that enters water from a known source.

Pollution: The release of harmful substances into the air, water, or land.

Population: A group of organisms of the same species that live in the same area. For example, a desert will have populations of different species of lizards and cactus plants.

Population Density: A measure of the number of organisms that live in a given area. The population density of a city may be given as the number of people living in a square kilometer.

Precipitation: Any type of liquid or solid water that falls to Earth's surface, such as rain, snow, or hail.

Prime Meridian: An imaginary north-south line that divides the planet into the Eastern Hemisphere and the Western Hemisphere. The prime meridian passes through Greenwich, England.

Pyroclastic Flow: A dense cloud of superheated gasses and rock fragments that moves quickly downhill from an erupting volcano.

R

Radiation: The transfer of heat/energy through space by electromagnetic waves

Rain Shadow: An area on the downwind side of a mountain that gets less precipitation than the side that faces the wind.

Recrystallization: The process by which bonds between atoms in minerals break and re-form in new ways during metamorphism.

Recycling: The reusing of materials that people would otherwise throw away, such as paper, glass, plastics, and certain metals.

Relative Age: The age of an event or object in relation to other events or objects.

Relative Humidity: The comparison of the amount of water vapor in air with the maximum amount of water vapor that can be present in air at that temperature.

Renewable Resource: A natural resource that can be replaced in nature at about the same rate as it is used.

Revolution: The motion of one body around another, such as Earth in its orbit around the Sun; the time it takes an object to go around once.

Richter Scale: The scale that measures the strength or magnitude of an earthquake

Rift valley: A deep valley formed as tectonic plates move apart, such as along a mid-ocean ridge.

Riparian Buffer: A vegetated area near a stream which helps shade and partially protect the stream from the impact of adjacent land uses.

Rock: A naturally formed solid that is usually made up of one or more types of minerals.

Rock Cycle: The set of natural, repeating processes that form, change, break down, and re-form rocks.

S

Salinity: The measure of the amount of dissolved salt contained in water.

Salt Water: Water that contains dissolved salts and other minerals. Oceans consist of salt water.

Seafloor Spreading: The process by which molten material adds new oceanic crust to the ocean floor

Sediment: A naturally occurring material that is broken down by processes of weathering and erosion.

Sedimentary Rock: Rock formed as pieces of older rocks and other loose materials get pressed or cemented together or as dissolved minerals re-form and build up in layers.

Seismic Wave: The energy waves (vibrations) caused by an earthquake.

Seismograph: An instrument that constantly records ground movements.

Sleet: Small pellets of ice that form when rain passes through a layer of cold air and freezes before hitting the ground.

Smog: The combination of smoke and fog; a type of air pollution that occurs when sunlight causes unburnt fuels, fumes, and other gasses to react chemically, often seen as a brownish haze.

Soil Horizon: A soil layer with physical and chemical properties that differ from those of soil layers above or below it.

Soil Profile: The soil horizons in a specific location; a cross section of soil layers that displays all soil horizons.

Solar Cell: A device that converts the energy of sunlight into electrical energy.

Sonar: A system that uses underwater sound waves to measure distance and locate objects.

Species: A group of living things that are so closely related that they can breed with one another and produce offspring that can breed as well.

Spectrum: Radiation from a source separated into a range of wavelengths. 2. The range of colors that appears in a beam of visible light when it passes through a prism.

Storm Surge: A rapid rise in water level in a coastal area that occurs when a hurricane pushes a huge mass of ocean water, often leading to flooding and widespread destruction.

Streak: The color of a mineral powder left behind when a mineral is scraped across a surface; a method for classifying minerals.

Stress: The force applied by an object pressing on, pulling on, or pushing against another object.

Subduction: The process by which an oceanic tectonic plate sinks under another plate into Earth's mantle.

Subduction Zone: A convergent plate boundary where one plate sinks or subducts beneath the other

Sunspot: A darker spot on the photosphere of the Sun. A sunspot appears dark because it is cooler than the surrounding area.

Sustainable: A term that describes the management of certain natural resources so that they are not harmed or used up. Examples include maintaining clean groundwater and protecting topsoil from erosion.

T

Technology: The use of scientific knowledge to solve problems or engineer new products, tools, or processes.

Tectonic Plate: One of the large, moving pieces into which Earth's lithosphere is broken and which commonly carries both oceanic and continental crust.

Tectonics: The processes in which the motion of hot material under a crust changes the crust of a space body. Earth has a specific type of tectonics called plate tectonics.

Telescope: A device that gathers visible light or another form of electromagnetic radiation.

Theory: In science, a set of widely accepted explanations of observations and phenomena. A theory is a well-tested explanation that is consistent with all available evidence.

Theory of Plate Tectonics: A theory stating that Earth's lithosphere is broken into huge plates that move and change in size over time.

Thunder: The sound wave created by intensely heated air around a lightning bolt.

Thunderstorm: A storm with lightning and thunder formed from a cumulonimbus cloud

Tornado: A violently rotating column of air stretching from a cloud to the ground.

Transform Boundary: A boundary along which two tectonic plates scrape past each other, and crust is neither formed nor destroyed.

Trench: A deep, steep-sided depression formed along the edge of a subduction zone created as a result of the subduction of one plate under another.

Tropical Storm: A low-pressure system that starts in the tropics with winds of at least 65 kilometers per hour (40 mi/h) but less than 120 kilometers per hour (74 mi/h).

Tsunami: A water wave caused by an earthquake, volcanic eruption, or landslide turnover

U

Ultraviolet Radiation: Radiation of higher frequencies than visible light, which can cause sunburn and other types of damage.

Urban Heat Island: The warmer body of air over a city.

V

Variable: Any factor that can change in a controlled experiment, observation, or model.

Volcanism: The process of molten material moving from a space body's hot interior onto its surface.

Volcano: An opening in the crust through which molten rock, rock fragments, and hot gases erupt; a mountain built up from erupted materials.

Volume: An amount of three-dimensional space, often used to describe the space that an object takes up.

W

Watershed: All of the area drained from a particular area of land that all flows into one body of water.

Water Cycle: The continuous movement of water on Earth, through its atmosphere, and in the living things on Earth.

Weather: The condition of Earth's atmosphere at a particular time and place.

Weathering: The process by which natural forces break down rocks.

Wetland: A wet, swampy area that is often flooded with water. (p. 468)

Wind: The horizontal movement of air caused by differences in air pressure & temperature